
Sandi Baressi Šegota

dr. sc., univ. mag. ing. comp. — Assistant Professor, Faculty of Informatics, Juraj Dobrila University of Pula

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PROFILE

Dr. sc. Sandi Baressi Šegota, univ. mag. ing. comp., is an Assistant Professor at the [Faculty of Informatics, Juraj Dobrila University of Pula](#), where his research predominantly focuses on Artificial Intelligence and Machine Learning. He earned his Doctorate of Science in Electrical Engineering in 2025. After completing Secondary Technical School Pula in 2013 and obtaining the qualification of Electrical Technician, he graduated from the Faculty of Engineering, University of Rijeka, with a bachelor's degree (*cum laude*) in Computer Science in 2017 and a master's degree (*cum laude*) in the same field in 2019. He enrolled in doctoral studies at the Faculty of Engineering in Rijeka later that year.

In the private sector, he serves as a Lead Developer at [dAlagnostics LLC](#), which focuses on groundbreaking research in health informatics and early neurodegenerative diagnostics. He also coordinates the architectural strategy and software implementation for the skincare mobile platform startup "[LiberiqAI](#)", exploring agentic structures. He frequently engages in joint academic outreach, delivering lectures at various events.

Throughout his academic career, Dr. Baressi Šegota has coauthored over 50 papers indexed in the Web of Science Core Collection, 30% of which he authored as the first author. He has also coauthored over 25 conference papers, 6 extended abstracts, and 1 book chapter. According to Google Scholar, his publications have been cited more than 1,000 times in total. His scientific and professional achievements have been recognized with several awards, including the 2021 Award for Exceptional Contribution to the Development and Promotion of Technical Culture by the Community of Technical Culture Pula and the Best Paper Award in the field of Artificial Intelligence in Medical Science at the SICAAI conference in 2023.

Beyond his academic and research activities, Dr. Baressi Šegota is an active member of the Astronomical Society "Istra" Pula (ADIP), which he joined in 2008. He has served as a member of the directorate since 2021 and as General Secretary since 2025. He serves as a representative in multiple bodies, including the Croatian Astronomical Union and the Community of Technical Culture, and volunteers for projects such as STEM Games and Istra-STEM. He has also been a member of the Institute of Electrical and Electronics Engineers (IEEE) since 2021 and of the Global Meteor Network (GMN) since its inception.

EDUCATION

- **Ph.D. in Electrical Engineering (dr. sc.), 2025**
[Faculty of Engineering, University of Rijeka](#)
- **Master's degree in Computer Engineering (mag. ing. comp.), cum laude, 2019**
[Faculty of Engineering, University of Rijeka](#)
- **Bachelor's degree in Computer Engineering (univ. bacc. ing. comp.), cum laude, 2017**
[Faculty of Engineering, University of Rijeka](#)

EXPERIENCE

- **Assistant Professor, 2025 - present**
[Faculty of Informatics, Juraj Dobrila University of Pula](#)
- **Lead Developer, 2025 - present**
[dAlagnostics LLC](#) — Health informatics and early neurodegenerative diagnostics
- **Architecture and software implementation, 2025 - present**
[LiberiqAI](#) — Skincare mobile platform startup; agentic structures

- **Research Assistant**, 2022 - 2025
Faculty of Engineering, University of Rijeka
- **Expert Associate - Junior Researcher**, 2019 - 2022
University of Rijeka — CEKOM SmartCity.4DII and DATACROSS projects

SKILLS

- **Robotics**
 - Industrial and collaborative manipulators
 - Path planning and trajectory optimization
 - Simulation (RoboDK, CoppeliaSim)
- **Artificial Intelligence**
 - Regression modeling
 - Convolutional neural networks
 - Data synthetization
 - Model interpretability
- **Evolutionary Computing**
 - Genetic algorithms and evolutionary optimization

AWARDS

- **2023** — Best Paper Award, Artificial Intelligence in Medical Science, SICAAI conference
- **2021** — Award for Exceptional Contribution to the Development and Promotion of Technical Culture, Community of Technical Culture Pula

MOST CITED PUBLICATIONS

1704 citations · h-index 21 · i10-index 39 (Google Scholar, 2026-08-08)

Z Car, S Baressi Šegota, N Anđelić, I Lorencin, V Mrzljak (2020). [Modeling the Spread of COVID-19 Infection Using a Multilayer Perceptron](#). *Computational and Mathematical Methods in Medicine* 2020, 10, 2020 **279 citations**

S Baressi Šegota, N Anđelić, I Lorencin, M Saga, Z Car (2020). [Path planning optimization of six-degree-of-freedom robotic manipulators using evolutionary algorithms](#). *International journal of advanced robotic systems* 17 (2), 1729881420908076, 2020 **109 citations**

D Štifanić, J Musulin, A Miočević, S Baressi Šegota, R Šubić, Z Car (2020). [Impact of COVID-19 on Forecasting Stock Prices: An Integration of Stationary Wavelet Transform and Bidirectional Long Short-Term Memory](#). *Complexity* 2020 (1), 1846926, 2020 **95 citations**

S Baressi Šegota, N Anđelić, V Mrzljak, I Lorencin, I Kuric, Z Car (2021). [Utilization of multilayer perceptron for determining the inverse kinematics of an industrial robotic manipulator](#). *International Journal of Advanced Robotic Systems* 18 (4), 1729881420925283, 2021 **81 citations**

S Baressi Šegota, I Lorencin, N Anđelić, V Mrzljak, Z Car (2020). [Improvement of marine steam turbine conventional exergy analysis by neural network application](#). *Journal of Marine Science and Engineering* 8 (11), 884, 2020 **69 citations**

J Musulin, S Baressi Šegota, D Štifanić, I Lorencin, N Anđelić, T Šušteršič, ... (2021). [Application of artificial intelligence-based regression methods in the problem of COVID-19 spread prediction: a systematic review](#). *International journal of environmental research and public health* 18 (8), 4287, 2021 **63 citations**

V Medica-Viola, S Baressi Šegota, V Mrzljak, D Štifanić (2020). [Comparison of conventional and heat balance based energy analyses of steam turbine](#). *Pomorstvo* 34 (1), 74-85, 2020 **54 citations**

N Anđelić, S Baressi Šegota, I Lorencin, V Mrzljak, Z Car (2021). [Estimation of COVID-19 epidemic curves using genetic programming algorithm](#). *Health informatics journal* 27 (1), 1460458220976728, 2021 **51 citations**

I Lorencin, S Baressi Šegota, N Anđelić, V Mrzljak, T Čabov, J Španjol, ... (2021). [On urinary bladder cancer diagnosis: Utilization of deep convolutional generative adversarial networks for data augmentation](#). *Biology* 10 (3), 175, 2021 **48 citations**

I Lorencin, S Baressi Šegota, N Anđelić, A Blagojević, T Šušteršić, ... (2021). [Automatic Evaluation of the Lung Condition of COVID-19 Patients Using X-ray Images and Convolutional Neural Networks](#). *Journal of Personalized Medicine* 11 (1), 28, 2021 **46 citations**

TEACHING

- **Network Systems** — Lectures, Faculty of Informatics - Juraj Dobrila University of Pula
- **Operating Systems** — Lectures, Faculty of Informatics - Juraj Dobrila University of Pula
- **Robotics** — Lectures, Faculty of Informatics - Juraj Dobrila University of Pula
- **Internet of Things** — Lectures, Faculty of Informatics - Juraj Dobrila University of Pula
- **Application of Artificial Intelligence** — Laboratory and Auditory Exercises, Faculty of Engineering - University of Rijeka
- **Evolutionary robotics** — Laboratory and Auditory Exercises, Faculty of Engineering - University of Rijeka
- **Electronics** — Laboratory Exercises, Faculty of Engineering - University of Rijeka
- **Programming** — Laboratory Exercises, Faculty of Engineering - University of Rijeka
- **Basics of robotics** — Laboratory and Auditory Exercises, Faculty of Engineering - University of Rijeka
- **Robotics in industry** — Laboratory and Auditory Exercises, Faculty of Engineering - University of Rijeka

SELECTED TALKS

- **Health Informatics - what to teach to students**
Conference, Pula Gymnasium, Pula, Croatia, 2026
- **Evolutionary Computing in Electrical Engineering**
Lecture, Dante Alighieri Secondary School, Pula, Croatia, 2026
- **Asteroid hunting from Pula**
Lecture, Pula Observatory, Pula, Croatia, 2026
- **Understanding the Energy Expenditure Predictors of Collaborative Robots Using Statistical and Machine Learning Methods**
Conference, 2026 49th MIPRO ICT and Electronics Convention (MIPRO), Opatija, Croatia, 2026
- **To Boldly Go - Robots in Space**
Lecture, Pula Observatory, Pula, Croatia, 2026
- **Influence of Operator Physical Characteristics on Compliance With Collaborative Robot**
Conference, IHSI 2026, Florence, Italy, 2026
- **Interpretable Multimodal Framework for Assessing Cognitive Load and Stress in Collaborative Robot Environments**
Conference, IHSI 2026, Florence, Italy, 2026
- **Johann Palisa 1848-1925**
Talk, Dom hrvatskih branitelja, Pula, Croatia, 2025
- **Identification of Synthetic Data Source Points Using Data Similarity Indexes and Artificial Neural Networks**
Conference, Springer Nature, Switzerland, Switzerland, 2024
- **On differentiating synthetic and real data in medical applications**
Conference, University of Belgrade, Belgrade, Serbia, 2023
- **Preparation of Simplified Molecular Input Line Entry System Notation Datasets for use in Convolutional Neural Networks**
Conference, University of Kragujevac, Kragujevac, Serbia, 2021

SERVICE AND LEADERSHIP

- General Secretary of the Astronomical Society "Istra" Pula (ADIP) since 2025; member since 2008, member of the administrative board since 2021
- Representative in the Croatian Astronomical Union since 2025
- Representative in the assembly of the Community of Technical Culture Pula since 2025
- Member of the Institute of Electrical and Electronics Engineers (IEEE) since 2021
- Member of the Global Meteor Network (GMN) since its inception
- Volunteer with STEM Games and Istra-STEM